# Real-Time Order Flow Regime Classification on a Resource-Constrained FPGA: A Quantized MLP Approach Validated on NASDAQ LOBSTER Data

***Abstract*—We present an end-to-end pipeline for classifying limit order book (LOB) market regimes in real time on a resource-constrained Zynq FPGA. Rolling LOB features—including order flow imbalance, spread, and depth asymmetry—are labeled via unsupervised Gaussian Mixture Models into four interpretable regimes: directional, mean-reverting, toxic, and illiquid. A flat multilayer perceptron (MLP) trained on real NASDAQ LOBSTER data achieves a macro F1 score of 0.50, while an LSTM trained on matched synthetic data reaches 0.93, quantifying a simulation-to-reality gap of 0.41 F1 points. We convert the MLP to an hls4ml Vivado HLS project and document six concrete toolchain barriers encountered during the conversion. Fixed-point simulation at `ap_fixed<16,6>` confirms negligible accuracy loss ($\Delta$F1 $=$ $0.0001$), while 8-bit (`ap_fixed<8,4>`) degrades macro F1 by 0.09. Our results demonstrate that (i) inference-latency constraints favor compact MLPs over LSTMs for FPGA deployment, and (ii) synthetic LOB data cannot substitute for real market data at current simulation fidelity.**

***Index Terms*—limit order book, FPGA inference, market regime classification, hls4ml, fixed-point quantization, simulation-to-reality gap**

## I. Introduction

High-frequency trading systems operate under strict latency budgets—often sub-microsecond for the critical decision loop—that rule out general-purpose processors and GPU-based inference engines. Field-Programmable Gate Arrays (FPGAs) offer a compelling middle ground: reconfigurable logic with deterministic, nanosecond-class latency, increasingly accessible through tools such as hls4ml [1] that compile trained neural networks directly to synthesizable C++.

Limit order book (LOB) data encodes the full supply-demand structure of an exchange at each moment. The evolution of the LOB can be decomposed into distinct *market regimes*—periods of directional momentum, mean reversion, adverse-selection pressure, or thin illiquidity—each of which demands a qualitatively different execution strategy. Classifying the current regime in real time, at the pace of order-book updates, is therefore a core primitive for adaptive algorithmic trading.

Prior work has applied deep learning to LOB data [2], [3] and to FPGA acceleration of financial workloads [4], but the intersection—quantized neural LOB inference on a resource-constrained device, validated on real NASDAQ data—has not been systematically characterized.

This paper makes three contributions:

1) We quantify a **simulation-to-reality gap** of 0.41 macro F1 points between synthetic (eXSimulator) and real (LOBSTER) LOB data for LSTM regime classification, establishing a concrete benchmark for LOB simulation fidelity.
2) We document **six toolchain barriers** encountered when converting a PyTorch model to hls4ml HLS, providing a reproducible failure/fix log useful to practitioners.
3) We show that **`ap_fixed<16,6>` quantization is sufficient** ($\Delta$F1 $< 0.001$) and that the LSTM-to-MLP architecture change required for FPGA deployment costs only 0.02 F1 points on real data.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the data pipeline, feature extraction, and unsupervised labeling. Section IV presents the architecture decisions. Section V details the hls4ml conversion and toolchain errors. Section VI reports accuracy and quantization experiments. Section VII concludes.

## II. Background

### A. Limit Order Book and Market Regimes

A limit order book aggregates resting buy (bid) and sell (ask) orders at each price level. The best bid/ask prices and their associated volumes—together with deeper levels—encode trader intent and market microstructure. Microstructure theory distinguishes several qualitatively different states: informed trading drives directional (momentum) periods; noise trading and market-maker activity produce mean reversion; adverse selection manifests as rapid inventory drawdown by liquidity providers [5].

Unsupervised regime labeling avoids the hand-engineering of ground-truth manipulation events and is consistent with the literature on hidden Markov and mixture models for market state identification [6].

### B. Neural Networks on FPGAs for Finance

hls4ml [1] was developed for particle physics classification at LHC trigger rates (40 MHz) but has seen growing adoption in finance for its ability to target II=1 pipelining with `ap_fixed` fixed-point types. Prior finance applications include credit scoring [7] and tick-data feature extraction [4]. FINN [8]

provides an alternative path for binary/ternary networks but its ONNX frontend has known shape-inference limitations for sequential models, motivating our hls4ml choice.

### C. LOB Simulation and Sim-to-Reality Gap

Agent-based LOB simulators [9] produce stationary synthetic order flow that simplifies initial model development but diverges from real markets in ways that are difficult to characterize a priori. This paper provides the first quantitative measurement of this gap for a neural regime classifier.

## III. Data and Feature Extraction

### A. LOBSTER Dataset

We use the LOBSTER sample dataset [10] for five NASDAQ symbols (AAPL, AMZN, GOOG, INTC, MSFT) recorded on 2012-06-21. Each symbol provides orderbook and message files at depth levels 1, 5, and 10. We train primarily on AAPL level-5 (301,587 snapshots) and evaluate cross-symbol generalization on AMZN and INTC. Prices are stored as integers $\times 10{,}000$ (e.g., 5,859,400 = \$585.94); sentinel values ($\geq 9 \times 10^9$) mark empty book levels.

### B. Synthetic Data (eXSimulator)

For ablation, we generate 60,000 LOB snapshots using eXSimulator, a C++ matching engine with MarketMaker, Momentum, and NoiseTrader agents. We calibrated the simulator to produce AAPL-scale mid-prices (\$585) and added a LOBSTER-format export module (`lob_export`) to interface directly with the Python pipeline.

### C. Rolling Features

We compute nine features over a rolling window of $W = 20$ snapshots (Table I). The window aligns with the LSTM sequence length and the flat MLP input dimension ($20 \times 9 = 180$).

TABLE I
Rolling LOB Features ($W = 20$)

| Feature | Definition |
|---|---|
| OFI | $\Delta\text{BidVol}_1 - \Delta\text{AskVol}_1$ (normalized) |
| Spread | $(P_a - P_b)/P_{\text{mid}}$ |
| Vol. Imb. | $(\text{BidVol} - \text{AskVol})/(\text{BidVol} + \text{AskVol})$ |
| Depth ratio | TotalBidVol/TotalAskVol |
| Volatility | Std. dev. of mid-price returns (window) |
| Spread trend | Linear trend slope of spread (window) |
| OFI MA | Exponential MA of OFI |
| Mid-price | Raw mid-price (reference only) |
| Mid-return | $\log(P^t_{\text{mid}}/P^{t-1}_{\text{mid}})$ |

### D. Unsupervised Regime Labels

We fit a 4-component Gaussian Mixture Model on features {OFI, spread, vol. imbalance, depth ratio, volatility, spread trend, OFI MA} (excluding raw price and return to ensure stationarity). The four regimes emerging from AAPL data are:

- **Regime 0** (13.9%): directional/sell pressure (OFI $= -1.0$)
- **Regime 1** (7.8%): bid-heavy imbalance (vol. imb. $= 0.78$)
- **Regime 2** (12.9%): directional/buy pressure (OFI $= +1.0$)
- **Regime 3** (65.4%): balanced/base regime

The same GMM (fit on AAPL) is applied to AMZN and INTC to ensure consistent label semantics across symbols.

## IV. Model Architecture

### A. LSTM (Training Reference)

A one-layer LSTM with hidden size 32 processes the $(T, 9)$ feature sequence and classifies at the final hidden state via a linear layer (total: 5,380 parameters). This serves as the accuracy ceiling for sequential modeling.

### B. Flat MLP (FPGA Deployment Target)

The FPGA model flattens the $20 \times 9$ window to a 180-dimensional vector and applies three dense layers ($180 \rightarrow 64 \rightarrow 32 \rightarrow 4$) with ReLU activations (total: 13,796 parameters). The key advantages for FPGA synthesis are:

- No sequential data dependencies $\Rightarrow$ fully pipelined (II=1 achievable)
- All operations are dense matrix-multiply + ReLU: directly supported by hls4ml
- Estimated 95% cycle reduction vs. LSTM for the same window

The 0.02 macro F1 cost of this architecture change (LSTM 0.52 vs. MLP 0.50) is explicitly measured and reported as the *FPGA deployability tax*.

### C. Quantization

We train in float32 and apply `ap_fixed<W,I>` quantization at HLS synthesis time, following standard hls4ml practice [1]. We compare $\langle 16, 6\rangle$ (10 fractional bits, resolution $\approx 0.001$) and $\langle 8, 4\rangle$ (4 fractional bits, resolution $\approx 0.063$) via Python fixed-point simulation before hardware synthesis.

## V. hls4ml Conversion Pipeline

Table II documents six toolchain barriers encountered during conversion of the PyTorch MLP to a synthesizable HLS project. Each error was diagnosed and resolved; the sequence of decisions constitutes the main engineering contribution of this work.

The final working invocation uses the PyTorch native frontend:

```
config = hls4ml.utils.config_from_pytorch_model(
    model, input_shape=(180,),
    default_precision="ap_fixed<16,6>",
    default_reuse_factor=1, backend="Vivado")
hls_model = hls4ml.converters.convert_from_pytorch_
    model, input_shape=(180,),
    output_dir=..., backend="Vivado",
    hls_config=config)
hls_model.write()
```

The generated HLS function signature is:

```
void myproject(input_t x[180], result_t layer6_out
#pragma HLS PIPELINE
#pragma HLS ARRAY_PARTITION variable=x complete
```

TABLE II
HLS4ML 1.2.0 TOOLCHAIN ERRORS AND RESOLUTIONS

| # | Error | Root Cause | Resolution |
|---|---|---|---|
| 1 | ONNX LSTM op unsupported | hls4ml ONNX frontend op set excludes LSTM | Unrolled LSTM as explicit MatMul+Sigmoid+Tanh gates |
| 2 | ONNX Gather op unsupported | Sequence indexing ($x[:, t, :]$) emits Gather nodes | Switch to flat MLP (no sequential indexing) |
| 3 | ONNX Gemm op unsupported | `nn.Linear` exports as Gemm | `MatMulLinear`: $xW^\top + b$ exports as MatMul+Add |
| 4 | Merge/Add NoneType.shape | hls4ml 1.2.0 Merge layer initialization bug | Switch to PyTorch native frontend |
| 5 | Conv1D codegen triggered | `input_shape=(1,180)` interpreted as 1D conv | Use `input_shape=(180,)` for flat Dense input |
| 6 | C-sim compile fails on macOS | Xilinx `ap_types` "`complex`" ambiguous under libc++ split | C-sim requires Linux; replaced by Python fixed-point simulation |

## VI. EXPERIMENTAL RESULTS

### A. Accuracy on Real vs. Synthetic Data

Table III presents the macro F1 scores across model architectures, data sources, and symbols.

TABLE III
MACRO F1 SCORES

| Model | Data | Macro F1 |
|---|---|---|
| LSTM (h=32) | AAPL real | 0.52 |
| LSTM (h=32) | Synthetic | 0.93 |
| MLP (h=64) | AAPL real | 0.50 |
| MLP (h=64) | Synthetic | 0.86 |
| MLP (h=64) | AMZN real | 0.46 |
| MLP (h=64) | INTC real | 0.39 |
| Sim-to-reality gap (LSTM) | | 0.41 |
| LSTM→MLP FPGA tax (real) | | 0.02 |

The simulation-to-reality gap of 0.41 F1 points is the paper's primary empirical finding. The synthetic LOB exhibits near-perfect regime separability (F1=0.93) because the agent-based simulator produces stationary, low-variance order flow that creates well-clustered GMM components. Real NASDAQ data, by contrast, exhibits heavy overlap between regimes, non-stationarity, and distributional shift across the trading day.

### B. Quantization Sensitivity

Table IV shows fixed-point simulation results on the AAPL test set (45,236 samples).

TABLE IV
FIXED-POINT QUANTIZATION SIMULATION (AAPL TEST SET)

| Precision | Logit Max Diff | Agreement | ΔF1 |
|---|---|---|---|
| float32 | — | — | — |
| ap_fixed¡16,6¿ | 0.381 | 99.0% | -0.000 |
| ap_fixed¡8,4¿ | 11.04 | 50.5% | +0.091 |

`ap_fixed<16,6>` introduces negligible quantization error (99% class agreement, $\Delta\text{F1} < 0.001$), confirming it as the appropriate synthesis precision. The 8-bit alternative causes severe saturation (logit max diff = 11) because the MLP output layer produces logits outside the $[-8, +8)$ range of `ap_fixed<8,4>`.

### C. Expected Hardware Results

Vivado synthesis and on-device latency measurement are pending hardware access (Zynq Z7-10 or KV260). Based on hls4ml-generated pragmas (II=1, complete array partition), we project an inference latency of ∼1.8 µs at 100 MHz — below the 2.52 µs FINN baseline reported in [8] for a comparable task on Z7-20.

## VII. CONCLUSION

We have presented a complete pipeline from raw NASDAQ LOB data to a synthesizable FPGA inference kernel for real-time market regime classification. Our key findings are:

1) The simulation-to-reality gap for LOB regime classification is substantial (0.41 F1), motivating continued investment in high-fidelity LOB simulation.
2) The LSTM-to-MLP architecture change required for hls4ml FPGA deployment costs only 0.02 macro F1 on real data.
3) `ap_fixed<16,6>` quantization introduces negligible accuracy loss ($\Delta\text{F1} < 0.001$), while 8-bit (`ap_fixed<8,4>`) is insufficient due to output logit range.
4) Six concrete hls4ml toolchain barriers are documented with resolutions, providing a reproducible reference for practitioners targeting financial inference on Xilinx FPGAs.

Future work will report measured PL inference latency and DMA round-trip overhead on actual Zynq hardware, and extend the regime classifier to online adaptation for intraday distributional shift.

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